Assignment 1 MHA RAKIB

17241003

1) $y(n) = x(n)\, u(n-10)$

For linear system, it should obey laws of additivity and homogneity.

$y_1(n) = x_1(n)\, u(n-10)$ — (i)

$y_2(n) = x_2(n)\, u(n-10)$ — (ii)

(i)+(ii) $= x_1(n)\, u(n-10) + x_2(n)\, u(n-10)$

Considering

$x_1(n) \rightarrow y_1(n) = x_1(n)\, u(n-10)$ — (iii)

$x_2(n) \rightarrow y_2(n) = x_2(n)\, u(n-10)$ — (iv)

(iii)+(iv) $= x_1(n)\, u(n-10) + x_2(n)\, u(n-10)$

multiplying constant with output

$= k(x(n)\, u(n-10))$

Similarly multiplying constant with input,

$= k(x(n)\, u(n-10))$

$\therefore$ the system is linear

Time variant

For a system to be TIV: —

-> no time scaling
-> coefficient should be constant
-> added term should be constant or zero.

delaying output by n_1

$\therefore y(n-n_1) = x(n-n_1)u(n-n_1-10)$

delaying input by n_1,

$= x(n-n_1)u(n-10)$

since coefficient is not constant, system is time variant.

Causality

$y(n) = x(n)\, u(n-10)$

the system does not depent on future values, thus causal.

stability.

B i/p $\rightarrow$ stable system $\rightarrow$ B o/p

B i/p $\rightarrow$ unstable system $\rightarrow$ U o/p

$x[n]$ = finite $\Rightarrow$ $y(n) =$ (finite) $u(n-10)$

thus unstable